

Article

A Comprehensive Analysis of Energy Consumption in Battery-Electric Buses Using Experimental Data: Impact of Driver Behavior, Route Characteristics, and Environmental Conditions

Mattia Belloni ^{*,†} , Davide Tarsitano [†]  and Edoardo Sabbioni [†] 

Department of Mechanical Engineering, Politecnico di Milano, 20156 Milan, Italy;
davide.tarsitano@polimi.it (D.T.); edoardo.sabbioni@polimi.it (E.S.)

* Correspondence: mattia1.belloni@polimi.it

[†] These authors contributed equally to this work.

Abstract: With the increasing emphasis on environmental sustainability, the electrification of urban public bus fleets has gained significant attention. Understanding the factors influencing the energy consumption of battery-electric buses (BEBs) is crucial for enhancing their energy efficiency. Therefore, it is crucial to identify the subsystems that contribute most to energy consumption and understand how operational factors influence them. This paper presents a comprehensive analysis of BEB energy consumption based on experimental measurements performed with a 12 m fully electric battery bus. The main limitations of this study stem from the use of a single vehicle over a total period of 18 days, during which 187 routes were completed. Additionally, sandbags were used as ballast in place of actual passengers. Various parameters, including the number of passengers, drivers, route characteristics, environmental conditions, and traffic, were analyzed to assess their impact on BEB energy consumption. Data related to the energy consumed by various bus utilities were collected through the vehicle's CAN network, with a sampling rate of 1 measurement per second. These data were analyzed both daily and per route, revealing the breakdown of energy consumption among different utilities and highlighting those responsible for the highest energy use. The results correlate the total distance traveled, service duration, average speed, driver's driving style, route characteristics, internal and external temperatures, and air-conditioning system's reference temperature with the energy consumption of the traction motors and climate control system. In addition, the correlation between the driver, vehicle acceleration, and throttle pedal use, and the energy consumed by the electric traction motor is presented.

Keywords: electric bus; energy consumption; driver characterization



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1. Introduction

The quality of air in cities is heavily influenced by the presence of vehicles that utilize the combustion of fossil fuels and locally emit large quantities of pollutant particles. To address this issue, public transport companies in the most densely populated cities have recently started to introduce electric buses into their fleets. This way, it is possible to use the energy produced from renewable sources or at least relocate the production of pollutant gases, thereby improving air quality within urban areas. To enhance energy efficiency in public bus transport, it is crucial to gain a comprehensive understanding of how these vehicles operate. Specifically, it is necessary to comprehend how the available electrical energy is used and in what proportions. Furthermore, the vehicle's energy consumption

is influenced by environmental factors, such as external temperature, traffic conditions, route, and number of stops. Lastly, the driver's behavior can significantly impact overall energy consumption. Even in the scientific literature, this topic has gained considerable attention [1–4]. The authors of [1] discussed the environmental benefits of fully electric buses in public transportation, providing novel empirical guidance for policymakers considering investments in electric urban buses. They evaluated the environmental advantages of using electric buses rather than diesel ones for urban transit. The authors of [2] reviewed a broad array of scientific literature on electric bus research, utilizing scientific mapping methods and content analysis to foster critical thinking. Their work highlighted the primary research streams, methodologies, and existing gaps in the field. The analysis suggests that future research on electric buses will predominantly focus on sustainability (including economic, environmental, and service quality dimensions), energy management strategies, and fleet operations. The authors of [3] analyzed the benefits of deploying electric bus fleets in China compared to diesel buses for public transportation. In their study, three electric buses from China were operated and charged in Macao under various air-conditioning, load, and speed settings. The study found that, at low speeds, with higher loads and air-conditioning usage, diesel buses showed greater improvements in on-road energy efficiency performance compared to electric buses. However, electric buses demonstrated superior energy and CO₂ emission benefits under these conditions. Overall, electric buses reduced petroleum use by 85–87% compared to diesel buses and achieved a 32–46% reduction in fossil fuel use and a 19–35% reduction in CO₂ emissions from a life-cycle perspective. Additionally, a cleaner power grid and an increase in system charging efficiency (if above 60–84%) would further enhance the future benefits of electric buses. To study the energy consumption of electric buses, several approaches have been adopted. The first method is a modeling approach, in which the components of the bus are modeled from an energy perspective. The second method primarily relies on experimental data, either acquired by simulating the working conditions of an electric bus or directly taken from vehicles in service. The third approach is a hybrid of the previous two approaches, where the results from the models are adjusted using experimentally measured data. The limitation of the first approach is related to the variability with which external disturbances affect consumption and how these are difficult to model. The second approach has the disadvantage of having to manage a large amount of data. The third approach is the most commonly used in the literature and appears to be the most promising. A fusion model integrating physical and data-driven approaches was developed by the authors of [5] to estimate the energy consumption of electric buses. The basic energy consumption of the electric bus is represented by a simplified physical model that considers the effects of rolling resistance, brake energy consumption, and air-conditioning usage. The model also accounts for the variability in energy consumption due to various factors. To predict the energy demand, the authors of [6] presented a novel approach to predict energy demand variation considering a wide range of uncertain factors. First, a factor identification process was conducted to recognize the variability in operating conditions. A computationally efficient surrogate model was then generated based on a previously developed numerical simulation model. The authors of [7] introduced a novel and versatile model for calculating electric bus energy consumption without requiring high-resolution speed profile data. The model generates a set of speed profiles using basic trip information: trip time, trip length, and distances between successive bus stops. The generated speed profiles accurately reflect various traffic conditions and speed behaviors encountered in real-world situations. Pan et al. [8] proposed a novel hybrid cumulative energy consumption prediction approach by combining vehicle dynamics with the forgetting factor recursive least squares method to achieve accurate real-time online forecasting. This approach was developed using real-world data

collected from over 983,000 frames across six electric buses in Guangzhou, China. In [9], the authors proposed an energy model for battery-electric buses, developed using established longitudinal dynamics models and event-based low-frequency data. The model accurately estimates energy consumption with limited input requirements and includes a comprehensive auxiliary systems model (heating, ventilation, air-conditioning, and other electrical components), which significantly contributes to the total energy consumption. The model was evaluated using data collected from electric buses in China and Norway. The authors of [10] focused on detecting and quantifying the correlations between the kinematic parameters of the vehicle and its energy consumption using real-world data. They used multiple linear regression to construct three energy consumption calculation models based on the vehicle dynamics equation. Each model utilized a different level of input parameter aggregation, allowing for predictions with various types of available input data. The authors of [11] presented an energy consumption prediction method for electric vehicles aimed at energy-efficient routing. This data-driven approach combines real-world driving data with geographical and weather information to predict the energy consumption for any road within a network. The study in [12] employed gray relational analysis to examine various external factors affecting the power consumption of electric buses and developed a feature library for similar days. Subsequently, they used a neural network to train the power consumption factors with data from this library, establishing a power consumption prediction model that incorporates multiple factors. In [13], the researchers evaluated the energy consumption and benefits of various hybrid and electric city bus powertrain layouts compared to conventional diesel powertrains using simulations. Their results indicated that plug-in hybrid and electric city buses have the greatest potential to reduce energy consumption and emissions. The authors of [14] offered a critical review, classification, and discussion of numerous state-of-the-art studies on the energy demand of electric buses. Ref. [15] provided a comprehensive review of the energy forecasting model, optimization objectives, charger technology, charging patterns, and constraints in electric bus scheduling. The authors of [16] examined additional energy consumption due to air-conditioning, proposing a method to evaluate extra energy consumption based on climate variables and planned public transport services. Their findings for numerous Italian provincial capitals and regions show that air-conditioning over-consumption is significantly impacted by daily mileage distribution, with considerable variations, even under similar climatic conditions. Lastly, ref. [17] assessed the energy consumption and battery performance of city transit electric buses operating on real routes and standardized bus drive cycles, using a framework that links bus electrification feasibility with real-world vehicle performance, transit service reliability, battery sizing, and charging infrastructure. Evaluating the driver's influence on an electric bus's energy consumption is crucial, as the operator significantly affects the motor's electrical consumption, a primary factor in overall energy usage. According to [18], data from 30 drivers over 46 days showed that road conditions and aggressive driving during rush hours heavily affect energy consumption. Ref. [19] utilized data-driven modeling to analyze how temperature, driver/vehicle, and road-related factors influence energy consumption using low-frequency real-world data. Additionally, ref. [20] investigated the driving parameters of 200 bus drivers, examining correlations with passengers, traffic, driving speed, acceleration, and driver drowsiness, and compared the impacts between pure electric buses and conventional fuel buses. The authors of [21] analyzed a unique real-world dataset to assist decision makers in the decarbonization of public fleets, while also considering the social acceptance of drivers. The study presented in [22] investigated the factors influencing the electricity consumption of electric buses under actual working conditions. The study was based on experimental measurements conducted in Belgrade over an extended period, covering typical seasonal

periods (winter, summer, and transitional) and specific daily operational periods (peak load, intermediate period, and first/last departures). The findings form the foundation for evaluating and ranking the influential factors affecting the electricity consumption of electric buses, which are crucial for planning line operations, analyzing energy efficiency, and determining operating conditions. The author of [23] proposed a novel method for estimating energy consumption on urban bus routes by analyzing driving and energy profiles. The energy consumption is derived from the travel parameters specific to the bus route, which are characterized by the driving profiles. The authors of [24] used an experimentally validated model to evaluate electric bus performance, including system efficiency and electric bus energy consumption. Finally, ref. [25] proposed a framework for predicting electric bus energy consumption, primarily divided into energy consumption estimation, kinematic feature prediction, and energy consumption prediction. The framework begins by fusing high-resolution real-world electric bus data with weather and road information, from which five types of influencing factors are extracted for different driving distances.

Given the growing interest in battery-electric buses, there is a lack of studies in the literature identifying the correlations between the energy consumption of the bus's component systems and its operating parameters. Most of the analyses in the existing literature rely on data derived from numerical models, which do not come directly from on-field experiments. While some research presents recorded instantaneous power data, the frequency of data collection is often insufficient for accurately estimating overall consumption, or detailed energy consumption data for individual utilities is absent. This study focuses on analyzing data collected during an experimental campaign aimed at identifying correlations between environmental factors and the subsystems of a battery-electric bus. The strength of this research lies in its ability to provide detailed consumption data for each utility, enabling the correlation of the energy consumption of each subsystem with the environmental conditions under which the vehicle operates. This paper presents data collected during an experimental campaign conducted with an electric bus operating on a regular bus route, using a sampling frequency of one measurement per second, and correlates the energy consumption of each vehicle's electric system with environmental factors. The collected data are initially analyzed by grouping data obtained at the end of each testing day to identify the systems with the highest energy consumption. Subsequently, the data are analyzed by breaking down the route of each experimental day into segments. This allows for the identification of correlations between the energy consumption of individual utilities and environmental parameters, including ambient operating temperature, air-conditioning set-point temperature, internal vehicle temperature, route characteristics, vehicle speed, vehicle accelerations, and driver behavior. Once the systems responsible for the highest energy consumption are identified, the manufacturers and users of these vehicles can target efficiency improvements. The results show that the driver's driving style can influence up to 50% of the power consumption of the traction motor.

2. Materials and Methods

This paper is structured as follows. First, a description of the experimental setup and the physical quantities measured is provided. Then, the operational conditions under which the experiment was conducted are presented, specifying the environmental and climatic factors. The results are then presented, beginning with the daily experimental results. Once it is established that the traction motor accounts for most of the energy consumption, the influence of factors such as mass, traffic conditions, average speed, acceleration, driver, and driving style on the energy consumed by the electric bus is analyzed. The relationship between the consumption of the air-conditioning system and temperature is also analyzed. Finally, a brief discussion of the obtained results is presented. The vehicle subject to

experimentation is a 12-meter-long electric bus with two axles, in which the rear axle is the driving axle. In the configuration used, the vehicle can accommodate 24 seated and 36 standing passengers, for a total of 60 passengers plus the driver. The vehicle storage system consists of lithium-ion batteries. In the vehicle, there are various systems necessary for proper service operation, such as the pneumatic suspension compressor, lighting system, door-opening system, power-steering assistance, and climate control system for both heating and cooling. As specified later, the experimental tests took place in summer; therefore, only the air-conditioning system was used. The electric layout of the vehicle is illustrated in Figure 1.

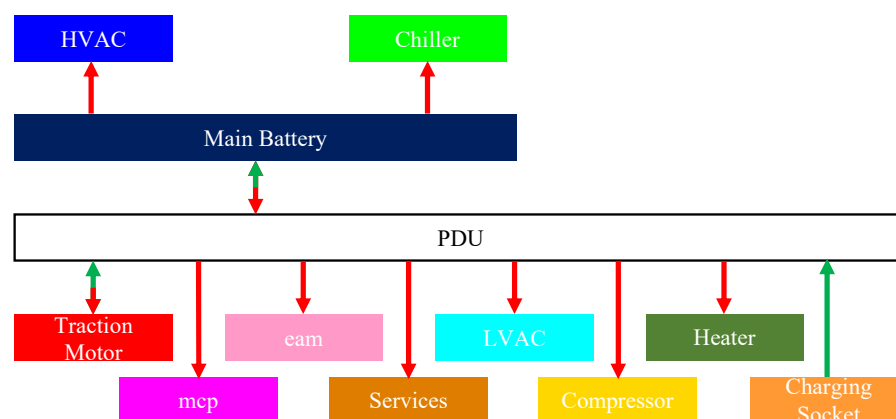


Figure 1. Electric scheme of battery-electric buses.

Green arrows indicate power flowing to the battery, while red arrows indicate power flowing from the battery. The main battery is connected to a power distribution unit (PDU) that manages the current drawn by the utilities and regulates power from the charging socket during recharging, as well as the electric motor driving system during regenerative braking maneuvers. Some utilities draw power directly from the main battery, such as the chiller for battery temperature regulation and the high-voltage air-conditioning system. In some cases, the data related to power consumption were already available from the peripherals and published on the CAN network. However, in most cases, power consumption was calculated by multiplying the instantaneous current and voltage values on the DC side. The data related to the absorbed current were provided by the control unit of the device to which it refers. These data are usually measured by a current sensor within the control unit or estimated by the control unit itself. The control unit also provides the voltage value, which is measured by a sensor inside it. The electric utilities of the battery-electric bus considered in this study are as follows:

- “Traction motor”: This includes the energy consumption of the AC traction motor and the inverter. The available data include the voltage and current on the DC side of the inverter.
- “mcp”: This refers to the motor cooling pump. The available data pertain to the voltage and the current absorbed. Although it is part of the 24 V circuit, it is essential for vehicle movement.
- “eam”: This is the electric power steering.
- “LVAC”: This is the air-conditioning system powered by the low-voltage circuit, including the ventilation fans and the driver’s cabin air-conditioning.
- “Services”: This refers to the power consumed by the 24 V circuit, excluding “MCP” and “LVAC”. This includes the power used by vehicle lights, door-opening mechanisms, and similar services.

- “Compressor”: This refers to the power consumed by the compressor in the suspension system.
- “Heater”: This is used to heat the vehicle cabin during winter but was not considered in this study due to summer measurements.
- “Chiller”: This is the battery pack chiller.
- “HVAC”: The high-voltage air-conditioning system cools the bus cabin. Like the battery chiller, this utility draws power directly from the battery pack.
- “Battery”: The current and voltage at the battery pack terminal were monitored to ensure the sum of energy consumed by all peripherals matched the power output from the battery pack.

Table 1 provides a detailed breakdown of how the power consumption of each device was calculated. Some control units provide an internally estimated power consumption value, while others supply current and voltage measurements at the power input. Unfortunately, the authors are unable to provide the specifications for each control unit.

Table 1. Data available on the CAN network from each peripheral.

Utility	Voltage	Current	Power
Traction motor	✓	✓	
mcp	✓	✓	
eam			✓
LVAC	✓	✓	
Services	✓	✓	
Compressor	✓	✓	
Heater			✓
Chiller			✓
HVAC			✓
Battery	✓	✓	

In addition to the previously mentioned values of power, current, and voltage, additional parameters were recorded to correlate the power consumption with the environmental conditions under which the vehicle operates. To monitor the vehicle’s dynamics, the position of the vehicle in GPS coordinates (latitude, longitude, and altitude), the odometer reading, the instantaneous vehicle speed, the position of the steering wheel, and the positions of the brake pedal and throttle pedal were recorded. The data were obtained from the vehicle CAN network according to the FMS standard with a sampling frequency of one measurement per second and transmitted to an FTP server using the Wi-Fi communication protocol. Then, the data were downloaded and analyzed at the end of the day.

The experimental campaign lasted 18 days. Data recording began when the bus started in the morning at the depot and continued until the bus returned to the depot in the evening. This approach ensured that the monitored consumption accounted for both the actual service periods, when the bus followed its scheduled routes, and the inevitable periods when the bus was in transit for necessary movements, such as positioning itself at the starting points of the routes or traveling to and from the depot at the beginning and end of the day. Energy consumption analysis was conducted by considering the total energy consumed throughout the entire day, while for individual routes, only the energy used from the departure of the first stop to the last stop of the same route was considered. The experimental campaign was conducted along four routes (R1, R2, R3, and R4) characterized

by different vertical slopes and lengths. Figure 2 presents the altitude profiles recorded during measurements as a function of the distance traveled by the bus from the bus stop. The four routes had different elevation profiles and lengths, so the results will be discussed separately for each route.

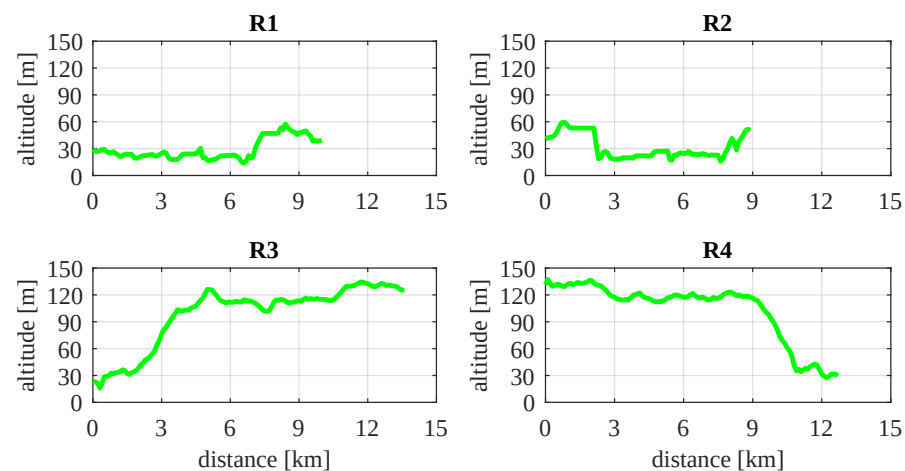


Figure 2. Altimetric profiles of the routes as a function of the distance traveled.

In Figure 2, each path is characterized by different vertical slopes, lengths, and numbers of stops, as summarized in Table 2. R3 has a significant elevation gain, while R4 shows a substantial negative elevation change. In contrast, R1 and R2 exhibit small variations in elevation along their routes. A comparison between Tables 2 and 3 reveals that, despite R1 and R2 being shorter routes, they were traveled more frequently, covering a total of 977.6 km. Conversely, R3 and R4, being longer routes, were traveled fewer times, covering a total of 1096.0 km.

Table 2. Route characteristics: length, vertical slope, and number of stops.

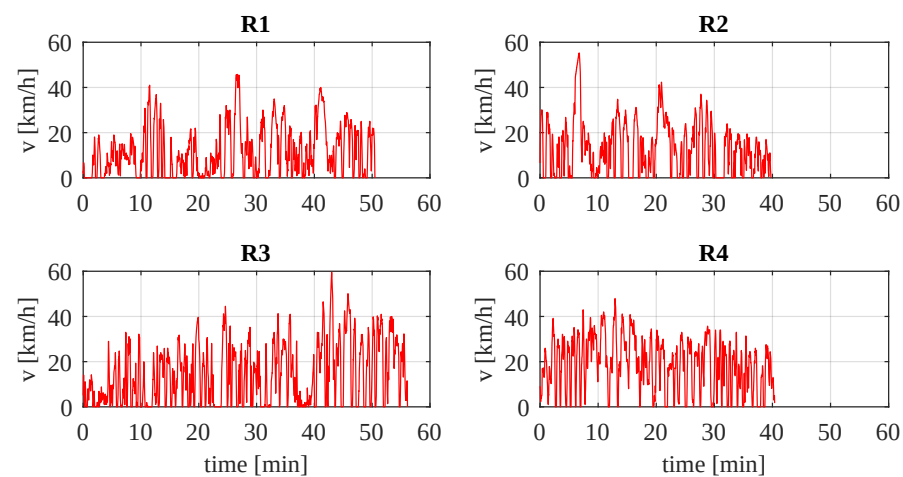
Route	Length [km]	Elevation Difference [m]	Number of Stops
R1	9.9	34	13
R2	8.9	−34	14
R3	13.6	114	17
R4	12.8	−114	14

During the experimental tests, the electric bus followed a conventional diesel bus on its regular route, ensuring that the doors opened at each bus stop as they would during normal service. A dedicated bus was used instead of an in-service bus to facilitate efficient monitoring of the additional load from onboard passengers. The impact of vehicle load was tested under three different conditions: full load (60 passengers), half load (30 passengers), and tare (no passengers). Each passenger was assumed to weigh 75 kg, and the unloaded vehicle mass was 13,540 kg. The data acquired during the experimental campaign included a total of 187 tracks. The correlation between driver behavior and energy consumption was investigated using data from 7 different drivers (D1, D2, D3, D4, D5, D6, and D7). Table 3 presents the number of bus routes each driver ran during the experimental campaign. The planning of the number of routes, the number of drivers involved in the experiment, and the bus loading conditions resulted from an agreement with the managers of the transportation service. So, driver rotation was maintained according to the availability of the transportation service company.

Table 3. Number of routes traveled per driver.

Driver	R1	R2	R3	R4	Total
D1	6	6	10	11	33
D2	12	12	10	11	45
D3	15	15	5	5	40
D4	6	6	3	2	23
D5	-	-	5	5	10
D6	9	9	3	2	23
D7	4	4	6	5	19
Total	52	52	42	41	187

The experimental campaign took place during different hours of the day, in different traffic conditions, and with different drivers. The vehicle speed was influenced by the environmental conditions under which it operated. As shown in Figure 3, the speed of the vehicle was influenced by external factors. Additionally, the vehicle speed dropped to zero at bus stops or in response to road signs. However, during the experimental campaign, the average vehicle speed ranged from 10 km/h to 25 km/h, as shown in Figure 3, with a maximum recorded speed of 70 km/h, in accordance with bus regulations.

**Figure 3.** Examples of the bus's speed profiles during service.

Before analyzing the energy consumption of the engine, air-conditioning, and other bus systems during service and examining their correlations with environmental and usage factors, it is important to understand how to visualize the data. In particular, the absolute value of the energy consumed is not very meaningful but is typically compared to the distance traveled by vehicles. However, referring to Figure 4, it is interesting to note that the distance traveled depends on the scheduling of the bus routes for a particular day, while the time taken to complete the entire service remains constant (around 13 h per day). This suggests that evaluating the performance of the battery pack in an electric bus should be based on the hours of service it provides, rather than the distance it can cover. With diesel buses, range anxiety was addressed by using a larger fuel tank, which had little impact on the total cost of the vehicle. In contrast, this becomes a crucial issue for battery-electric buses, where the cost of the batteries significantly affects the overall cost of the vehicle. Furthermore, at least during the initial stages of the transition from a diesel fleet to an electric fleet, it is quite common for fleet management purposes and to limit the

charging infrastructure costs, to plan for a single overnight battery charge, using the energy accumulated throughout the day.

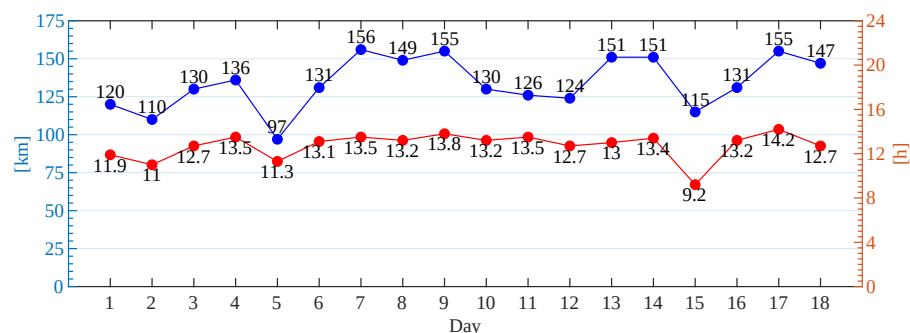


Figure 4. Distance traveled and time required to complete the daily service.

As mentioned earlier, the experiment took place during the summer, when temperatures were very high, necessitating significant use of the air-conditioning system. Figure 5 shows the variations in the average temperature and the set temperature of the air-conditioning system during the experiment. It can be observed that the average temperature was around 34.5 °C, while the set temperature of the air-conditioning system varied depending on the user.

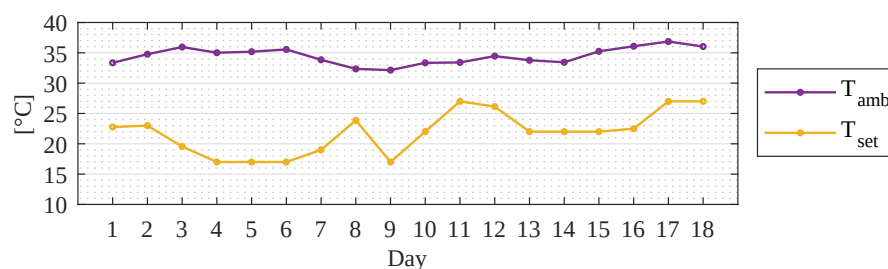


Figure 5. External and set-point temperatures of the bus.

3. Results

The previous section outlined the conditions under which the experiment was conducted. This section presents the results obtained in detail. First, the impact of each system on total energy consumption is shown. Then, the main systems, namely the traction motor and air-conditioning system, are analyzed in relation to their dependency on the following environmental factors:

- Vehicle mass;
- Route characteristics (particularly elevation changes);
- Traffic;
- Average vehicle speed;
- Vehicle acceleration;
- Driver;
- Driving style;
- Temperature.

Figure 6 presents a breakdown of the bus's energy consumption, expressed as a percentage of total daily energy usage, distributed across individual systems. The systems with the highest energy consumption were the traction motor (mot 44.91%) and the air-conditioning system (HVAC 28.72% and LVAC 5.62%). The third-largest energy consumer, accounting for approximately 11.17% of daily consumption, was the power steering system (eam). The remaining components, listed in descending order of consumption, included

the compressor (comp 6.78%), 24V system services (serv 3.81%), traction motor cooling pump (mcp 1.76%), and chiller (chil 0.2%).

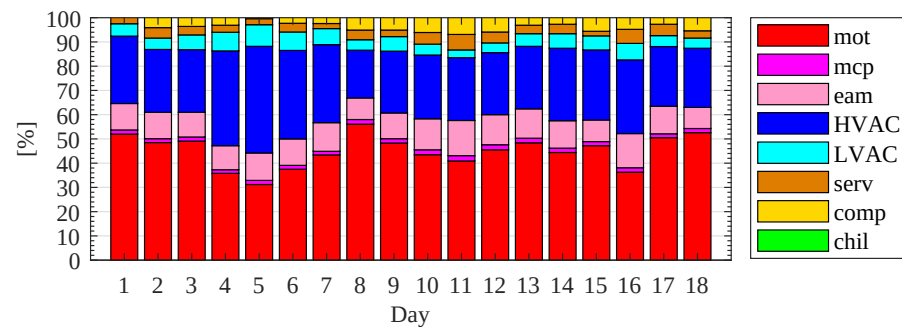


Figure 6. Energy consumption distribution of the systems of the electric bus.

The energy consumed by the traction motor was the largest contributor to the vehicle's overall energy consumption. The variability in consumption was due to various external factors (e.g., route characteristics and driver behavior), which are analyzed later. Figure 7 shows the variation in the ratio between the energy regenerated by the traction motor during braking phases and the energy consumed during traction phases ($e_{mot} = \frac{E_{reg}}{E_{cons}} \cdot 100$), as a function of the day of the experiment. This ratio averaged 54.5%.

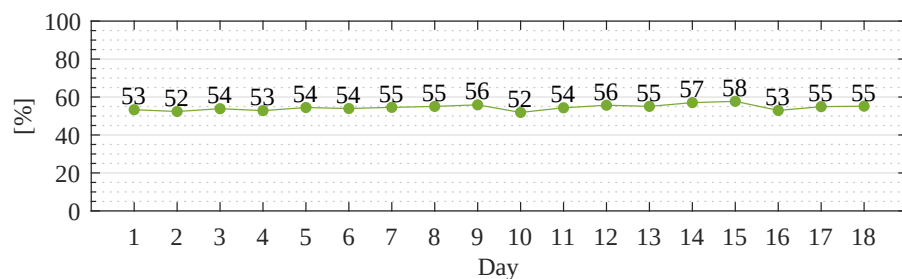


Figure 7. Ratio of the energy regenerated to the energy consumed by the traction motor as a function of the day of the experiment.

To evaluate the performance of the traction motor, it is essential to analyze energy consumption using a parameter that is not directly influenced by the distance traveled or the vehicle's mass. It is intuitive that absolute energy consumption increases with the distance traveled. Figure 8 shows that the different vehicle loading conditions (full load, half load, and empty) influenced energy consumption. For each route, the energy required by the bus to travel 100 km decreased as the vehicle's mass decreased. To eliminate dependency on the vehicle's mass, similar to the approach taken for distance traveled, energy consumption per kilometer can be normalized by the vehicle's mass. Therefore, from this point onward, the energy consumption of the electric motor is expressed in $\frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}$, representing the energy used to move one ton of mass over 100 km. It is important to highlight the variation in average consumption for each route. R1 and R2 exhibited similar average consumption rates ($4.0 \frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}$ and $3.0 \frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}$, respectively). R3, characterized by a significant positive elevation gain, exhibited a much higher average consumption than R4 ($6.0 \frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}$ versus $1.3 \frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}$).

To assess the impact of traffic conditions, Figure 9 shows the trip durations based on departure times. Across all four routes, typical variations in trip duration were observed during peak hours, from 9 to 11 a.m. and again from 5 to 7 p.m. However, the fluctuations in durations remained relatively moderate, suggesting that traffic patterns throughout the day do not vary significantly, except for routes R1 and R2, where a notably shorter duration

(about 50% less than the average trip time) was observed. The average trip duration for route R1 was 43 min; for R2, 36 min; for R3, 48 min; and for R4, 43 min.

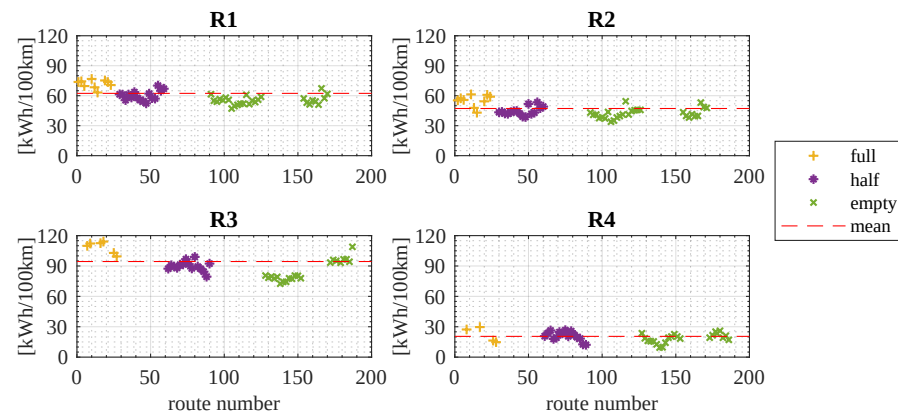


Figure 8. Influence of vehicle mass on traction motor energy consumption.

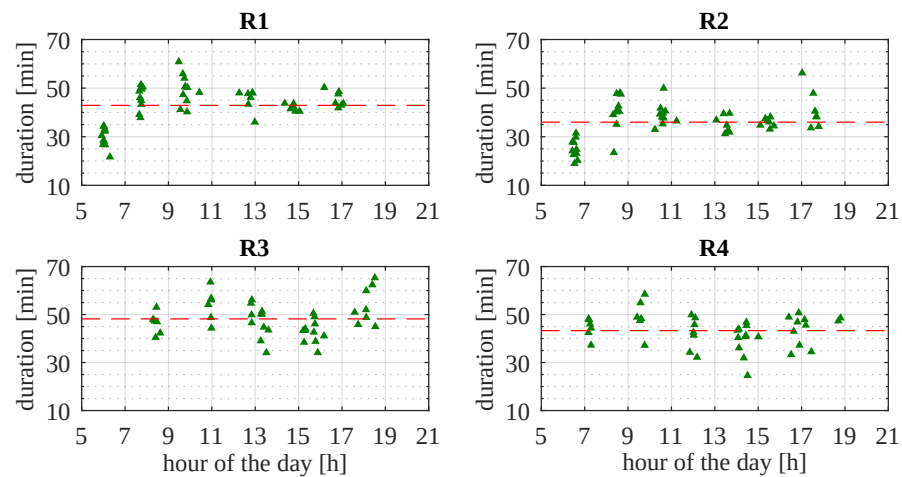


Figure 9. Trip duration as a function of starting time. The dotted lines represent the average value of trip duration.

As shown in Figure 10, the energy consumption, normalized to the distance traveled and vehicle mass (expressed in $[\frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}]$), was unaffected by trip duration. This is because the electric motor of the bus in the experiment did not consume energy when the vehicle was stationary in traffic, with the vehicle speed being zero. However, unlike the traction motor, an analysis of the energy consumption of all auxiliary systems revealed a relationship between the energy consumed and the trip duration. Figure 11 shows the ratio of the energy consumed by the auxiliary systems (including all vehicle systems, except for the traction motor) to the total energy consumed on that trip as a function of trip duration. A clear upward trend in energy consumption with increasing trip duration was observed. Considering Figures 6, 10 and 11, it can be concluded that the energy consumption of the auxiliary systems, particularly the air-conditioning system—which alone accounted for 34.34%—increased with service duration, ultimately contributing more than 100% additional consumption to the total auxiliary energy usage of the vehicle. This means that the energy costs for the same transport service double during peak hours.

Figure 12 illustrates the relationship between the average speed recorded during a trip and the energy consumption of the traction motor. As expected, there was a positive correlation between energy consumption and average speed. However, this trend exhibited significant variability, making it difficult to assess the vehicle's energy consumption solely based on the average speed maintained during the trip.

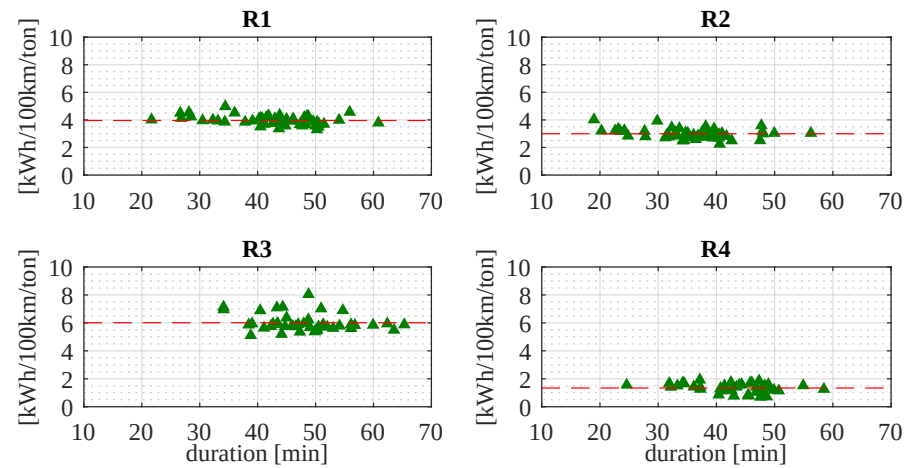


Figure 10. Energy consumed by the electric bus as a function of trip duration. The dotted lines represent the average value of the specific energy consumed.

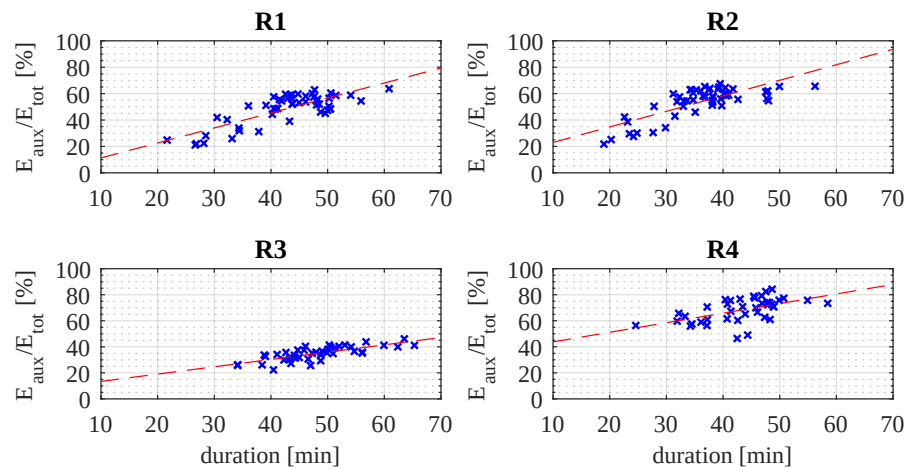


Figure 11. Ratio of energy consumed by auxiliary systems to total energy as a function of trip duration. The dotted lines represent the trend of the energy consumption ratio.

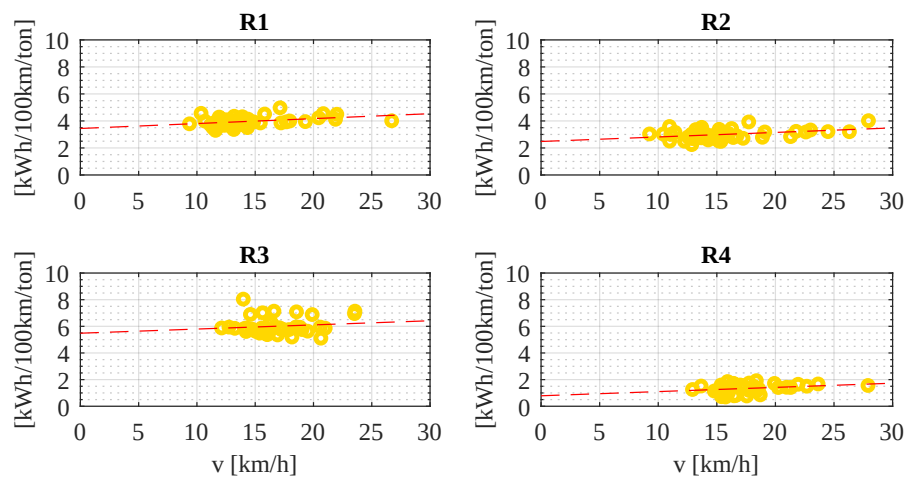


Figure 12. Energy consumption of electric traction motor as a function of average vehicle speed. The dotted lines represent the trend of average vehicle speed.

Unlike traditional diesel buses, electrically powered buses can regenerate some of the vehicle's kinetic energy by redirecting power back to the battery pack. Figure 13 illustrates the relationship between vehicle accelerations and energy consumption/regeneration. To distinguish driving style and the average level of acceleration and deceleration during a

trip, and thus correlate them with the energy consumed/regenerated by the vehicle, an acceleration index was introduced. The index represents the percentage of instances where vehicle acceleration exceeded 0.5 m/s^2 (or fell below -0.5 m/s^2) relative to the total duration of the route. This threshold of 0.5 m/s^2 signifies an abrupt maneuver, considering that a bus's driving style must guarantee a certain comfort level for the passengers. The graph demonstrates that motor consumption increased with sudden accelerations. Conversely, a hard braking maneuver resulted in more regenerated energy.

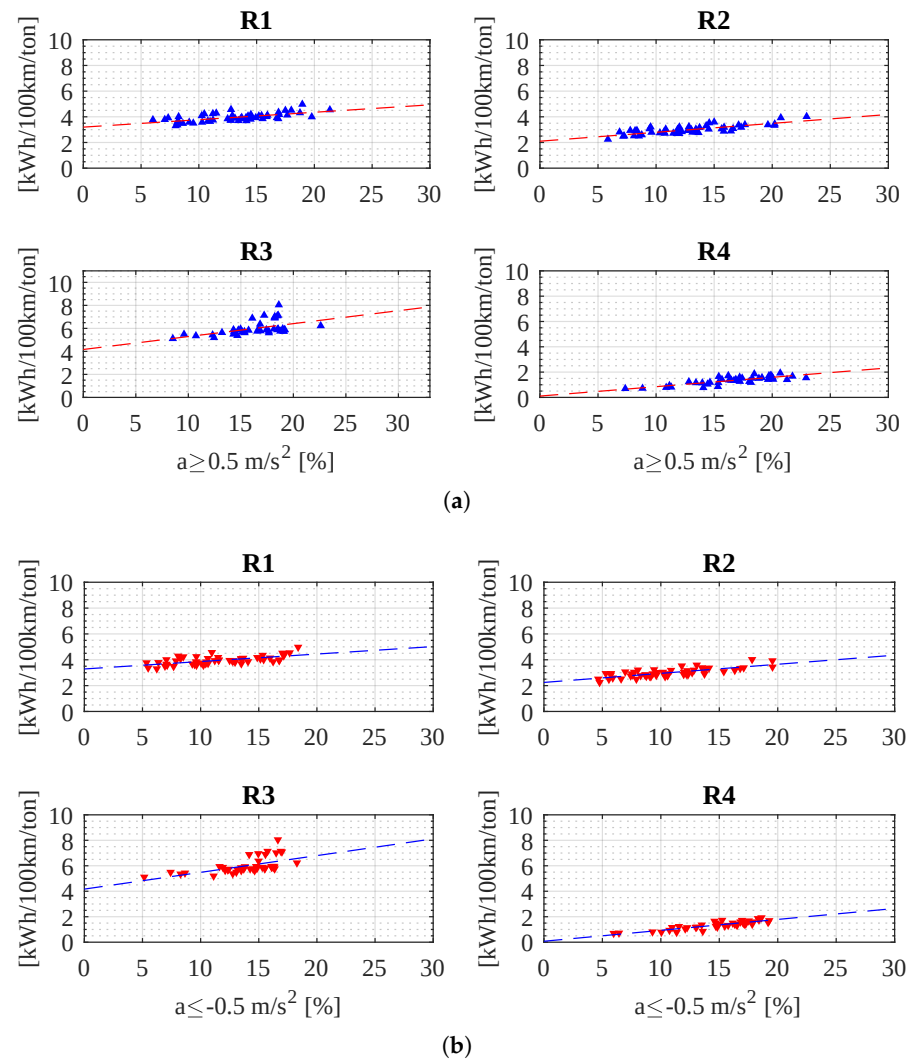


Figure 13. Energy consumption of the traction motor during the traction phase (a) and regenerated energy during braking maneuvers (b). The dotted lines represent the trend of motor energy consumption as a function of the acceleration ratio.

To investigate the influence of the driver on the energy consumption of the traction motor, Figure 14 depicts the mean values and standard deviations of energy consumption normalized by distance and vehicle mass $[\frac{\text{kWh}}{100 \text{ km} \cdot \text{ton}}]$. Drivers D5 and D3 were the most efficient in utilizing the electric motor's potential, consistently demonstrating lower energy consumption compared to their peers. In contrast, drivers D2 and D7 generally consumed more energy than average. These findings suggest a clear correlation between driving style and the performance of the electric motor. To highlight the extremes, when comparing driver D3 to driver D7, on route R3, D7's consumption was approximately 31% higher, and on route R4, it peaked at a 100% increase. However, it is important to consider that these observations of individual driving styles were based on a statistical sample, as shown in Table 3, which may not be large enough to draw definitive conclusions. Additionally,

not all drivers covered the same number of routes. It is noteworthy that drivers who traveled longer distances tended to exhibit lower energy consumption (e.g., D1 and D3). This suggests the presence of a learning curve, where drivers gradually adapt to operating an electric motor and optimize its performance over time.

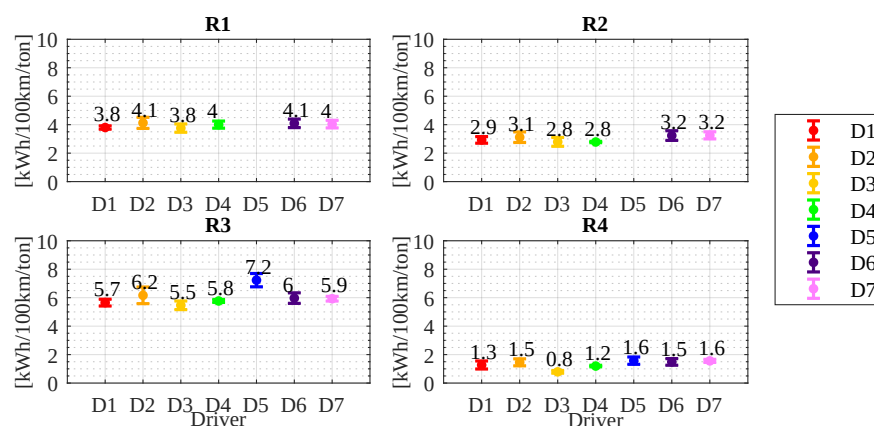


Figure 14. Influence of drivers on traction motor energy consumption.

The correlation between energy consumption and acceleration was analyzed by considering the vehicle's dynamics. However, the torque transmitted to the ground by the electric motor (which causes vehicle acceleration/deceleration) is correlated with the accelerator pedal demand. The accelerator serves as the human-machine interface through which the driver controls the vehicle. In modern buses, the correlation between accelerator pedal demand and the torque delivered by the motor results from a complex layering of control logic that maintains vehicle maneuverability; thus, they are not directly correlated. It is possible to describe the driver's behavior based on their action on the accelerator pedal. The degree of accelerator pedal opening can be expressed as a percentage (0–100%). Similar to the analysis of accelerations, driving style can be assessed by examining the amount of time during the route when a specific percentage of throttle is applied. Figure 15 illustrates the correlation between the energy consumption of the traction motor and the fraction of time the driver was acting on the throttle pedal. The throttle request is expressed as the ratio of the instances in which the driver pushes a certain level of the throttle pedal to the total number of instances in which the driver is pushing the throttle pedal. Figure 15a shows the driver requesting less than 25% throttle, while Figure 15b shows the correlations when the driver requested more than 50% throttle. From the first graph, it is possible to assess how cautiously a driver requested energy from the motor, while the second graph provides insights into when the driver consistently requested larger amounts of power from the motor. The results in Figure 15a show that drivers D1 and D3 consumed less energy and pressed the throttle pedal less than 25% for a longer time. Conversely, drivers D5 and D6 used the throttle pedal, requiring more than 25% for a more significant amount of time. These results are confirmed by Figure 15b. Drivers D1 and D3 requested more than 50% throttle for a shorter duration compared to drivers D5 and D7, and, consequently, their mean energy consumption was lower than that of their less thrifty colleagues. The conclusion is that drivers who apply greater pressure to the accelerator pedal exhibit higher traction motor energy consumption. Therefore, a more “aggressive” driving style results in increased energy consumption.

The ballast used consisted of sandbags, which offer several advantages, such as precise load monitoring. However, a limitation of this method is that it does not accurately replicate the thermal capacity of typical bus passengers. During the experimental campaign, the bus doors were opened at each bus stop to simulate the real-world operating conditions of a bus in service. This allowed the hourly climate control consumption to be related to

the difference between the ambient temperature and the target temperature set by the air-conditioning system, as shown in Figure 16. The correlation between energy consumption and the temperature difference is clearly evident: energy consumption increased as the difference between the external temperature and the AC system's set temperature increased. The chart reflects the average power consumption recorded during the bus's operation on its routes, excluding breaks between runs and transfers from depots.

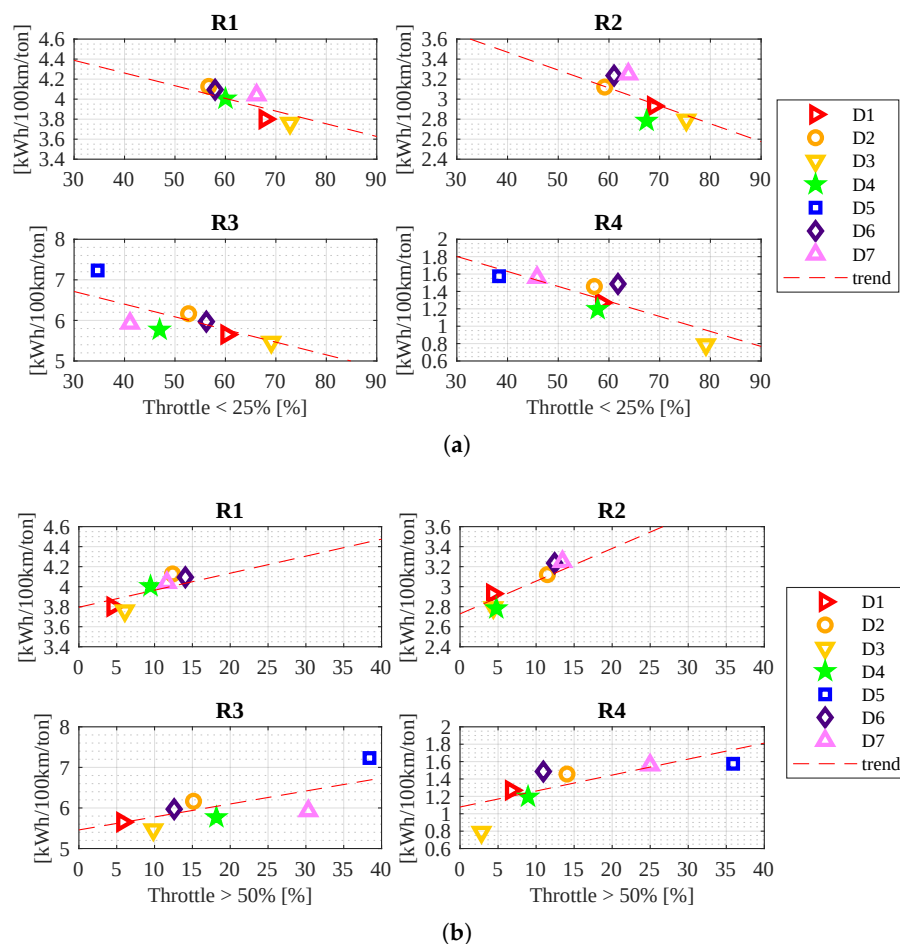


Figure 15. Influence of throttle pedal use on bus energy consumption. (a) Energy-saving driving style. (b) Inefficient driving style.

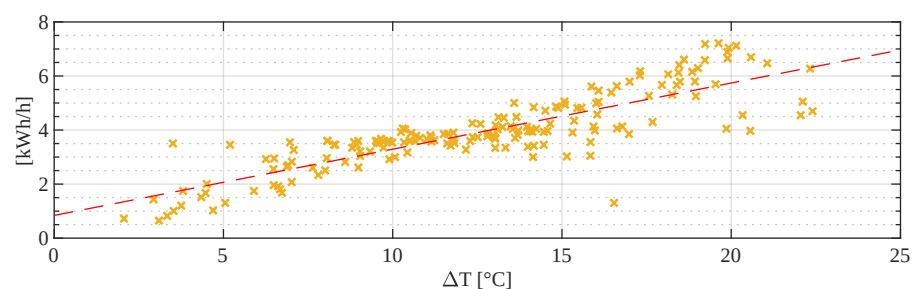


Figure 16. Air-conditioning system energy consumption. The dotted lines represent the trend of the energy consumed as a function of the temperature.

4. Discussion

The previous two sections presented the results of the experiment conducted on a 12-meter battery-electric bus. First, the operational conditions under which the experiment took place were described, followed by an analysis of the relationship between energy consumption and environmental factors. The first key observation was that the traction

motor accounted for the majority of the vehicle's energy consumption. However, by closely monitoring the ratio of the traction motor energy consumption to the other systems, it was evident that the electric motor allowed for efficient use of the energy stored in the battery pack. This was made possible by the motor's ability to recover a portion of the vehicle's kinetic energy and return it to the battery. The results show that up to 50% of the energy consumed could be regenerated, unlike a propulsion system using an internal combustion engine. Next, the relationship between the energy consumption of the traction motor and various operational factors was investigated. Initially, a strong correlation was observed between the route's altimetric profile and energy consumption, as the latter was affected by the significant mass of the vehicle. Furthermore, the results showed a relationship between energy consumption and vehicle load conditions. For the experiment, sandbags were used as ballast to monitor the additional weight on the vehicle. However, the thermal capacity of public transport users was not accurately replicated. Despite this approximation, it did not limit the experiment's objectives, as it allowed for a more precise relationship to be established between the vehicle mass and the energy consumption of the traction motor. The traffic conditions in which the vehicle operated were also analyzed. The traffic pattern followed typical peak-hour trends, with two main congestion periods: from 9 to 11 a.m. and from 5 to 7 p.m. The energy consumption of the electric motor was not affected by the time taken to cover the same route, as the motor consumed no energy when the vehicle was stationary, unlike buses with internal combustion engines. However, a clear relationship was found between auxiliary system consumption and trip duration, indicating that traffic conditions have an overall impact on total energy consumption. Regarding the relationship between the average speed and energy consumption of the traction motor, no net correlation was found. Although energy consumption increased with higher average speed, the low dispersion of data points for average speed during trips meant that this relationship was not significant for evaluating the bus's energy consumption. In contrast, a much stronger relationship was observed between the energy consumed/recaptured during acceleration/braking phases. An index was proposed to represent the ratio of instances in which the vehicle exceeded 0.5 m/s^2 of acceleration/deceleration to the total instances recorded during a trip. This index provided valuable insight into the vehicle's dynamics and the driver's driving style. This relationship was further explored by examining the mean and variance of the energy consumption of the traction motor for each driver. The results indicated that under certain conditions, there could be up to 50% energy savings between a driver who is less mindful of consumption and one who is more energy-efficient. Furthermore, analyzing the correlation between the recorded energy consumption and the number of trips completed by each driver suggested a learning curve, with more experienced drivers demonstrating lower energy consumption. However, due to limited data, this relationship was not explored further. Instead, the relationship between driver input and the energy consumption of the traction motor was analyzed more thoroughly. Similar to the analysis of accelerations, an index was proposed to assess the driver's request for torque at the accelerator pedal. The first index counted instances when the driver requested less than 25% throttle, while the second counted instances when the driver requested more than 50% throttle. The results aligned with those of the previous analysis, where more energy-conscious drivers tended to request less throttle. Finally, the relationship between the energy consumption of the air-conditioning system and the difference between the external temperature and its set-point temperature was examined. A strong trend was observed between the temperature difference and hourly energy consumption, with peaks showing up to three times the required power as the temperature difference ranged from 5 to 20 °C.

The analysis of the results presented so far focuses on the periods when the electric bus was actually in service, excluding movements to and from the depot, as well as transfers required to reach the starting points of the routes. It is interesting to examine the breakdown of energy consumption between the moments when the bus was actively in service and those when it was engaged in preparatory maneuvers before actual service. Figure 17 illustrates the total energy consumption during the experiment, broken down by utilities and further categorized into the periods when the vehicle was actively providing service along the route and “out-of-service” periods. The primary sources of energy consumption were the electric motor and the air-conditioning system. Of the total energy used, 84.88% was attributed to the traction motor during active service, while the air-conditioning system accounted for up to 33.6% of total energy consumption during the experimental campaign.

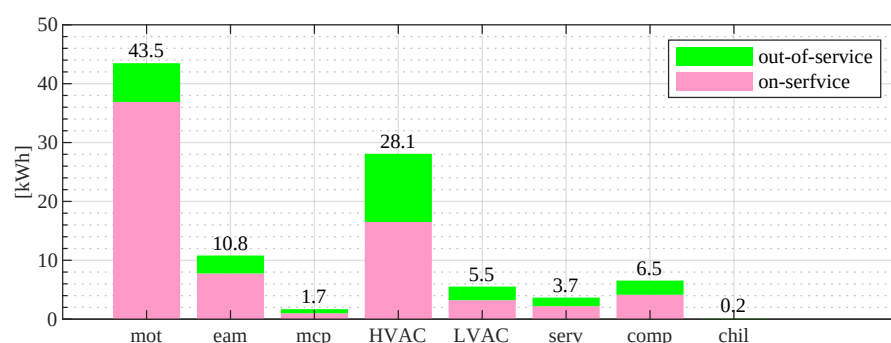


Figure 17. Repartition of the total energy consumed during the experimental tests.

This statistic, combined with the previous analysis, prompts reflection on the overall energy consumption of vehicles used in public transportation. The technology implemented in electric-powered vehicles offers advantages in terms of energy consumption and reductions in local pollutant emissions. However, it must be used appropriately. During this experiment, drivers were instructed to behave as they would when driving a traditional bus powered by an internal combustion engine. The limited energy availability provided by the battery pack, however, requires more mindful use of resources. Alongside improvements in the efficiency of the vehicle’s installed systems, users must develop awareness to help optimize energy savings. Possible solutions could include better management of the air-conditioning system or more efficient accelerator use, allowing drivers to synchronize their driving with road infrastructure, such as traffic lights. Furthermore, the planning of routes by the public transport operator, combined with optimized depot management and charging infrastructure, could contribute to additional energy savings. Finally, the use of dedicated bus lanes could help reduce travel times, leading to significant improvements in both service quality and energy efficiency.

5. Conclusions and Future Developments

The majority of analyses in the existing literature are based on data from numerical models rather than direct field experiments. Although some studies provide recorded instantaneous power data, the sampling frequency is often too low to accurately estimate total energy consumption. Additionally, detailed insights into the energy consumption of individual utilities are often lacking. This article aims to bridge the gap identified in the scientific literature regarding the assessment of energy consumption in electric vehicles for public transport. While most studies primarily rely on numerical models, this research takes a different approach by conducting an experimental study that replicates real operating conditions. This paper presents the results of experimental tests aimed at correlating the energy consumption of vehicle subsystems with operating parameters. The analyses

conducted confirm the numerical data on energy consumption derived from the traction motor while also enabling a more accurate measurement of the distribution of other consumption factors. The results highlight the dominance of the traction motor in the vehicle's overall energy consumption while also showcasing the efficiency of the electric motor in regenerating energy through kinetic recovery. Significant correlations were observed between energy consumption and factors such as the altimetric profile of the route, vehicle load, and traffic conditions. Although traffic does not directly affect the electric motor's energy consumption, it impacts the energy consumption of auxiliary systems. Additionally, while no strong correlation was found between average speed and energy consumption, a clear relationship was observed between energy consumption/regeneration and acceleration/braking phases, underlining the influence of driving style on energy efficiency. An index based on accelerator usage was proposed to evaluate driver influence. Based on this index, it was determined that a driver specifically trained for electric bus operation could reduce energy consumption by approximately 50% compared to an untrained driver.

This study was limited to a single bus model, tested during a specific season and on a limited number of routes. To overcome these limitations, future research could analyze different seasonal conditions and a broader range of routes and vehicles.

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